



Creating AI-Agents with Transformer Model and MCP to Provide Treatment/Diagnosis of Common Illnesses

Dennis Tislin

Computer Systems Lab -Yilmaz- 2026



Background/Problem

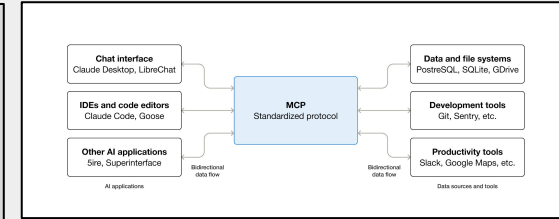
The cost of technology and treatment within healthcare has been continuously increasing over time –Medicare spending grew 8.1% to \$1,029.8 billion in 2023 and Medicaid spending grew 7.9% to \$871.7 billion in 2023. With this continuous increase, the availability of treatment for common illnesses has decreased for lower/middle income population. In order to ameliorate this exacerbating issue it is needed that treatment becomes accessible with low costs, especially for cases regarding common illnesses, occurring often. By creating a cost-efficient, reliable A.I. integrated alternative to illness treatment and diagnosis, the flawed healthcare system seen today can eventually be improved. Through the utilization of Large Language Models (LLM) on Hugging Face and fine tuning an optimized domain-specific model, an A.I. Agent consisting of a Model Context Protocol (MCP) Client, LLM, and MCP Server can replicate the basic tasks of medical workers who provide diagnosis, treatments, and prescription suggestions of common illnesses.

System Architecture

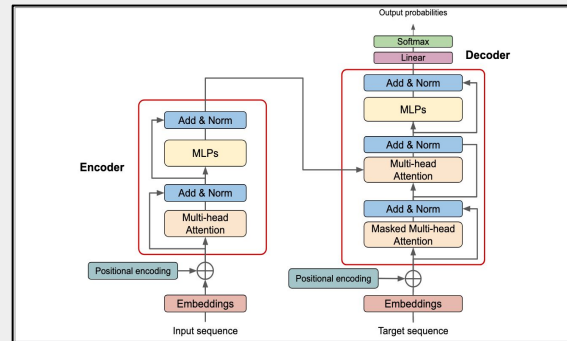
By utilizing Transformer Models to process sequential patterns and understand the relationships of English in Natural Language Processing, an A.I. Agent connected to a Model Context Protocol server can effectively diagnose the symptoms of illnesses, suggest treatment, and contact medical providers for over the counter medicine/treatment. By processing user input into the model to predict the best output, the model provides an accurate diagnosis. On a command fulfilling a condition, the Model determined if the MCP Client must connect to the server and execute an action from a pipeline consisting of tools.

Results

When fine tuning the Llama3 Open Bio LLM on text data containing over 5000 medical records of common illnesses using Pytorch and Hugging Face libraries, several epochs of tuning Low Rank Adapters produced a quantized model with ~0.021 training and validation loss, preventing overfitting and eliminating bias. After creating the MCP server tools using Asyncio and the Server library along with necessary access tokens required for accessing/sending information, a JSON configuration was added within the Claude Desktop MCP Client to successfully integrate the fine tuned model and server. With this, the A.I. Agent was successfully implemented to function in a conversational environment. When connecting to the server pipeline, the LLM successfully determined which commands/actions were necessary for execution based on the input. The server was able to consistently connect and execute Web Search tasks, along with sending diagnosis information through Gmail.



```
{
  "mcpServers": {
    "memory": {
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-memory"]
    }
  }
}
```



Conclusion/Future Work

While successfully being able to diagnose common illnesses, uncertainty is still present; an output prediction of multiple of illnesses exists. Also, an appropriate user interface was not yet implemented for the Agent, as Claude's ubiquitous interface was used because of its role of being the Client. This limitation will be addressed in the future, along with the broader expansion of the Agent's specialization; accounting for rare illnesses would further improve its utilization within healthcare. Other models, such as Buoy A.I., GPT 4.0, and DxGPT contain user-input limitations addressed within my project. utilizing Cloud Services with a larger model could further improve accuracy within the healthcare domain. Furthermore, the Claude Desktop MCP Client was utilized, because of its compatibility with MCP through JSON configuration. For a completely novel innovation, creating an original MCP Client would be necessary through the creation of an A.I. Agent Framework.



User
Prompt

MCP
Client